Deadline:

- **Deadlines: 11:59 PM Chicago time, on 11th November, 14th November, and 19th November, 2026 (Chicago Clock).**
- The submission link will be closed automatically after the deadline. **No submissions will be accepted after the deadline (including via email)**. Please have sufficient time before the deadline to scan your work and upload it to Canvas.

Submission Instructions:

- Total Points: 100
- Work in a group of three (3) students. We require one submission per group at each checkpoint. Please see the submission details at the end of the document.
- Submit at "Assignments ⟶ Projects ⟶ Project 2 Assignment Links"

Resource for MIPS MARS Simulator: Following are the resources for the MIPS MARS simulator that we will use for this project.

1. **MIPS MARS Simulator Download Link**: Click Here.
2. **Link for Java SDK**: Click Here.
3. **Behavioral Modeling**: Click Here.
4. **Tutorial on how to download and install MARS, all MARS features, how to assemble and execute code, and how to see the outputs and register/memory contents**: Click Here.

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1 Problem Sets

```
def fibonacci(n):
  if n <= 1:
    return n
  a = 0
  b = 1
  for i in range(n - 1):
    temp = b
    b = a + b
    a = temp
  return b
```

Figure 1: **Pseudo-code of Fibonacci computation**.

```
def IsOdd(m):
    remainder = 0
    remainder = division(m, 2)
    if remainder == 0:
        return False
    else:
        return True
```

Figure 2: **Pseudo-code of Odd number computation**.

```
def division(x,y):
    quotient = 0
    while x >= y:
        # repeat subtractions
        x = x - y
        quotient = quotient + 1  # update count of subtractions
    return x
```

Figure 3: **Pseudo-code of Division by Repeated Subtractions (DRS)**.

Rev: 10/28/2026

Problem 1 (Designing a Fibonacci Odd-Number Detection Program). In this problem set, we will design a MIPS program that will compute if the nth Fibonacci number is an odd number, *i.e.*, **IsFibonacciOdd**(n). To implement this program, we will design three main functions: (i) design a function **Fibonacci**(n) to compute n^{th} Fibonacci number, (ii) design a function **IsOdd**(m) to compute if a number m is odd, *i.e.*, when m is divided by 2, the remainder is 1, and (iii) design a top level function that will first compute n^{th} Fibonacci number and then use the computed number to check if it is odd.

(a) Design a MIPS program that will implement **Fibonacci**(n) of Figure 1. [Points **:** 50]

(b) Design a MIPS program that will implement **IsOdd**(m) of Figure 2. Please use **division by repeated subtraction** to implement $m\%2$ (reads m modulo 2 and computes the remainder of the division $m/2$. Assume that m is a nonnegative integer.). Usage of MIPS DIV instruction will yield a zero (**0**) point. Use the function of Figure 3 to implement the division by subtraction. [Points **:** 20]

(c) Design a MIPS Program **IsFibonacciOdd**(n) using the programs from Part (a) and Part (b). [Points **:** 30]

You can use any number of registers for computation. Assume that the input n is a nonnegative integer stored in memory. Retrieve n from memory, perform the required computation, and store the result in a different memory location. The result should be '1' if **True** and '0' if **False**.

2 Submission Deadlines, Expectations, and Artifacts

Create a GitHub repository for your group. The GitHub repo will be used for collaborative coding by your team members. Commit and push your codes as frequently as possible. The commit history may be used as an indication of the contribution of each of the team members. A nice GitHub cheat sheet is available here.

(A) **Progress Report 1**: Complete the progress report with the following details – (i) Assembly code (`.asm`) for **Fibonacci**(n) MIPS program (ii) A **README** file containing the name of the team members and their NetIDs, contribution of each team member, GitHub Link, and how we can run the MIPS program to get the nth Fibonacci number for a given n. Submit a tar ball containing the MIPS `asm` file and the **README** file.

- **Deadline**: **11/11/2026 at 11:59 pm Chicago Time**.
- **Expectation**: Completion of Fibonacci sequence computation.

(B) **Progress Report 2**: Complete the progress report with the following details – (i) Assembly code (`.asm`) for **IsOdd**(m) MIPS program (ii) A **README** file containing the name of the team members and their NetIDs, contribution of each team member, GitHub Link, and how we can run the MIPS program to find if a given number m is odd. Submit a tar ball containing the MIPS `asm` file and the **README** file.

- **Deadline**: **11/14/2026 at 11:59 pm Chicago Time**.
- **Expectation**: Completion of Odd number computation.

(C) **Final Report**: Complete the final report with the following details – (i) Assembly code (`.asm`) for **IsFibonacci Odd**(n) MIPS program (ii) A **README** file containing the name of the team members and their NetIDs, contribution of each team member, GitHub Link, and how we can run the MIPS program to find if n^{th} Fibonacci number is odd. Submit a tar ball containing the MIPS `asm` file and the **README** file. Tutorial on how to tar is available in the following link: (How to tar?). We may pull code from your GitHub, and please ensure that your submitted code and GitHub code match and are the latest versions.

- **Deadline**: **11/19/2026 at 11:59 pm Chicago Time**.
- **Expectation**: Completion of entire problem.

Rev: 10/28/2026

NO EXTENSION OF ANY DEADLINE WILL BE GRANTED UNDER ANY CIRCUMSTANCES.